Notes: Bertrand & Mullainathan (QJE, 2001)
Are CEOs Rewarded for Luck? The Ones Without Principals Are

Contents

1 Big picture

- **Question.** Are CEOs paid for changes in firm performance that are beyond their control? If so, does the extent of this pay for luck depend on the quality of corporate governance?
- **Contracting benchmark.** In a simple principal-agent model, shareholders should not reward a CEO for observable luck. If the board observes an external shock to performance, such as an oil price change, the optimal contract filters that shock out.
- **Central empirical fact.** CEOs are rewarded for luck. CEO compensation responds to observable exogenous shocks approximately as strongly as it responds to ordinary firm performance.
- **Three luck measures.** The paper uses oil prices for oil companies, industry-specific exchange-rate shocks for traded-goods firms, and industry mean performance for the broad Yermack CEO sample.
- **Skimming interpretation.** The authors propose a skimming view: in weakly governed firms, CEOs partly capture the pay-setting process and can increase compensation when good performance creates less shareholder scrutiny.
- **Governance test.** If pay for luck reflects skimming, better governed firms should filter luck more. The paper tests this using large shareholders, large shareholders on the board, CEO tenure, board size, and insider share.
- **Main governance result.** Better governed firms pay less for luck. The strongest evidence comes from large shareholders on the board. Adding a large shareholder on the board reduces pay for luck by about 23–33 percent.
- **Why this is important.** The result challenges a simple optimal-contracting view of CEO pay and gives evidence that principal-agent models work best when there are actual principals with power to monitor management.
- **Broader takeaway.** CEO pay is shaped by both incentive contracting and managerial power. Pay for performance is not automatically evidence of efficient incentives, because performance-based pay can include pay for observable luck.

2 Incremental contribution of the paper

- **From pay-performance sensitivity to pay-for-luck sensitivity.** Earlier CEO-pay studies emphasized whether pay rises with performance. This paper asks a sharper question: does pay rise with the component of performance that is clearly outside the CEO's control?
- **Observable luck as a diagnostic test.** The paper uses observable shocks as a test of whether compensation contracts filter information efficiently. This converts a broad debate about CEO incentives into a concrete empirical test.
- **Multiple luck measures.** The paper does not rely on one proxy. Oil price shocks, exchange-rate shocks, and industry-wide performance shocks each provide different forms of exogenous variation.
- **Separation of skimming from contracting stories.** Finding pay for luck alone does not prove skimming, because some extended contracting stories could justify pay for luck. The incremental contribution is to test whether pay for luck declines with governance quality.
- **Governance interaction design.** The paper studies whether governance affects pay for luck relative to general pay for performance. This is stronger than simply asking whether poorly governed firms pay more overall.
- **Direct evidence on managerial power.** The finding that better governed firms filter luck better provides evidence that CEO influence over pay-setting matters.

- **Implication for empirical corporate finance.** The paper warns that pay-performance sensitivity can be a misleading statistic. If pay responds to luck, then a positive pay-performance correlation may not reflect incentive provision.

3 Data and measurement

3.1 CEO compensation data

- The broad CEO data come from David Yermack's compensation and governance data set.
- The sample contains large publicly traded U.S. firms from 1984–1991.
- Firms enter the sample if they appear frequently in Forbes 500 rankings by sales, profits, assets, or market value.
- Total compensation includes salary, bonus, and the value of options granted in the year.
- The authors exclude changes in the value of options already held, because unindexed option portfolios mechanically move with stock-price luck.
- Salary and bonus are also analyzed separately because they are discretionary and therefore particularly informative about board filtering.

Interpretation. The exclusion of accumulated option-value changes is conservative. If those changes were included, pay for luck would be even larger because unindexed options mechanically reward stock-price luck.

3.2 Oil-company sample

- The oil sample covers 50 of the largest U.S. oil firms from 1977–1994.
- The key luck variable is the real price of crude oil.
- Oil prices are a useful luck measure because they are observable, economically large, and plausibly outside the control of individual CEOs.
- The oil application tests whether CEO pay in a sector with obvious external shocks rises with those shocks.

Interpretation. The oil sample is a clean case study because the external shock is easy to identify and visualize. Figures I–III show that oil prices, oil industry performance, and oil CEO pay move together.

3.3 Exchange-rate luck

- For traded-goods firms, exchange-rate shocks affect firm performance depending on the firm's industry exposure.
- The paper uses exchange-rate movements as instruments for performance.
- Exchange-rate shocks are useful because they are external to the CEO and vary across industries depending on import/export exposure.
- The exchange-rate test helps show that pay for luck is not an oil-industry-specific phenomenon.

3.4 Industry-performance luck

- The broadest luck measure is the asset-weighted average performance of other firms in the same two-digit industry.
- For accounting performance, the instrument is industry operating income to assets, excluding the firm itself.
- For market performance, the instrument is industry log shareholder wealth, excluding the firm itself.
- The industry-performance measure is close to relative performance evaluation, but the paper interprets it as luck: a sector-wide shock outside the control of an individual CEO.

Interpretation. The exclusion of the firm itself prevents mechanical reflection. The industry instrument captures sector fortunes rather than the CEO's own contribution.

3.5 Governance measures

- **Large shareholders:** number of 5 percent blockholders.
- **Large shareholders on board:** number of 5 percent blockholders who sit on the board.
- **CEO tenure:** used as a proxy for CEO entrenchment, especially when no large shareholder is present.
- **Board size:** larger boards are often weaker monitors.
- **Fraction insiders:** higher insider share can indicate weaker board independence.
- **Governance index:** standardized combination of governance variables.

Interpretation. The governance measures are imperfect but economically motivated. Large shareholders on the board are especially powerful because they combine cash-flow incentives with direct monitoring capacity.

4 Models and empirical specifications

4.1 Simple contracting benchmark

Let performance be

$$p = a + do + u,$$

where a is CEO action, o is observable luck, d is the effect of luck on performance, and u is unobservable noise. In the optimal Holmstrom-Milgrom-style contract, compensation filters observable luck:

$$s = \alpha + \beta(p - do) = \alpha + \beta(a + u).$$

Interpretation. Observable luck should not be rewarded because it does not improve incentives and only adds risk to compensation. This is the core theoretical benchmark.

4.2 Oil-industry pay-for-luck regression

The oil case study estimates specifications of the form

$$\ln(Comp_{it}) = \alpha_i + \gamma_t + \beta \ln(OilPrice_t) + X'_{it}\theta + \varepsilon_{it}.$$

The key coefficient is β.

Interpretation. If oil CEOs are rewarded for luck, their compensation rises when oil prices rise, even though no individual CEO controls crude oil prices.

4.3 General pay-performance regression

A baseline pay-performance equation is

$$y_{it} = \alpha_i + \gamma_t + \beta Perf_{it} + X_{it}'\theta + \varepsilon_{it},$$

where y_{it} is log total compensation and $Perf_{it}$ is either accounting performance or shareholder wealth.

Interpretation. This estimates ordinary pay-performance sensitivity, but it does not distinguish CEO-controlled performance from luck.

4.4 Pay-for-luck IV equation

The same equation is estimated by instrumenting firm performance with luck:

$$Perf_{it} = \pi_0 + \pi_1 Luck_{it} + controls + v_{it}.$$

Then the second-stage pay coefficient measures pay for luck.

Interpretation. Comparing OLS pay-performance sensitivity with IV pay-for-luck sensitivity asks whether CEOs are paid differently for lucky performance than for general performance.

4.5 Governance interaction model

The paper estimates

$$y_{it} = \alpha_i + \gamma_t + \beta Perf_{it} + \theta(Gov_{it} \times Perf_{it}) + \lambda Gov_{it} + X_{it}'\delta + \varepsilon_{it}.$$

For pay-for-luck regressions, both $Perf_{it}$ and $Gov_{it} \times Perf_{it}$ are instrumented with industry performance and its interaction with governance.

Interpretation. The interaction coefficient asks whether governance changes pay sensitivity. The key comparison is whether governance reduces pay for luck more than it reduces general pay for performance.

4.6 Tenure and entrenchment specification

The paper splits firms by whether a large shareholder sits on the board and then examines how pay for luck changes with CEO tenure.

Interpretation. The skimming view predicts that CEO tenure should increase pay for luck when no strong monitor is present, because long-tenured CEOs have more time to entrench themselves.

5 Identification and empirical design

5.1 What identifies pay for luck

The identification strategy uses observable shocks that affect firm performance but are plausibly outside the CEO's control. These include oil prices, exchange rates, and industry-wide performance. The paper then asks whether CEO pay responds to performance caused by those external shocks.

Interpretation. The logic is a filtering test. If boards can observe the shock, they should remove it from pay if the contracting model is correct.

5.2 Why three luck measures matter

- Oil prices provide a transparent industry case study.
- Exchange-rate shocks provide industry-specific external variation for traded-goods firms.
- Mean industry performance provides a broad measure usable for many firms and for governance interactions.

Interpretation. Agreement across all three measures strengthens the inference that pay for luck is a general feature of CEO pay, not a quirk of one industry or one instrument.

5.3 Why governance is central

Pay for luck alone could have multiple interpretations. It could reflect skimming, CEO labor-market outside options, incentives to forecast shocks, or inability to filter. The governance test helps distinguish these theories. If pay for luck were optimal or filtering were impossible, well-governed firms should not systematically filter luck better than poorly governed firms. But if pay for luck reflects skimming, better governance should reduce it.

5.4 Key empirical caveat

The governance results are not necessarily causal effects of changing governance. They are cross-sectional relationships. Firms with large shareholders or smaller boards may differ in other ways. The paper treats the governance findings as evidence for the relevance of the skimming view, not as a policy experiment proving exactly how to reform boards.

5.5 Remaining threats

- Luck measures may not be fully outside CEO influence in every context.
- Industry shocks may partly affect CEO outside options.
- Governance variables may proxy for firm size or other unobserved features.
- Compensation data exclude accumulated option-value changes.
- Governance findings are suggestive, not randomized.

6 Section-by-section study guide

- **Introduction.** Frames pay for luck as a test of contracting versus skimming.
- **Pay-for-luck test.** Develops the contracting benchmark and estimates pay for luck using oil prices, exchange rates, and industry performance.
- **Why is there pay for luck?** Discusses skimming, CEO labor-market explanations, forecasting incentives, and filtering-impossibility explanations.
- **Governance tests.** Shows that firms with stronger monitors pay CEOs less for luck.
- **Conclusion.** Argues that both contracting and skimming matter, and that principal-agent models work best when real principals are present.

7 Tables and figures

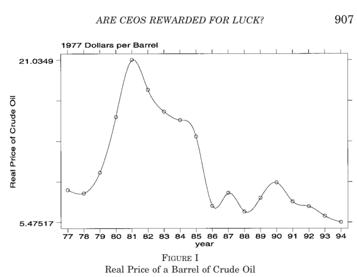
7.1 Figures I–III: Oil price, oil industry performance, and oil CEO pay

Read:

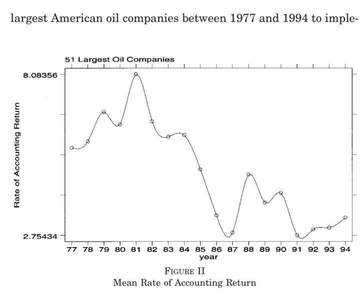
- Figure I plots the real price of crude oil.
- Figure II plots mean accounting returns in the oil industry.
- Figure III plots oil CEO pay and crude oil prices.
- The visual pattern suggests that oil-company performance and CEO pay move with oil prices.

Detailed interpretation. These figures motivate the oil case study. Oil price movements are large, visible, and outside the control of individual CEOs. If CEOs are rewarded when oil prices raise performance, that is a clean example of pay for observable luck.

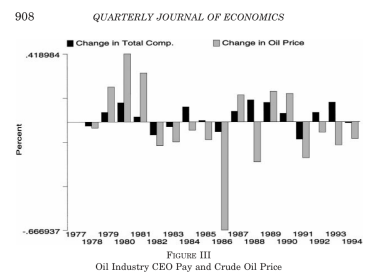
Economic message. The figures show why oil prices are a useful first test of whether CEO pay filters exogenous shocks.



(a) Figure I: Real price of crude oil



(b) Figure II: Oil accounting returns



(c) Figure III: Oil CEO pay and crude oil price

7.2 Table I: Pay for luck for oil CEOs

Read: Table I reports regressions of log total compensation on oil prices and performance. Oil CEOs are paid when crude oil prices rise. The effect appears in total compensation and in salary and bonus.

Detailed interpretation. The salary-and-bonus result is important because these are discretionary components of pay. If boards wished to filter oil-price luck, these components are where they could offset mechanical option gains. Instead, they also rise with luck.

Economic message. Oil CEOs are rewarded for a shock they do not control.

7.3 Table II: Summary statistics, full Yermack CEO sample

Read: Table II describes the broad CEO compensation and governance sample. Average total compensation exceeds salary and bonus because option grants are important. The average CEO is about 57 years old and has about nine years of tenure. The average firm has about 1.12 large shareholders and a board of about thirteen directors.

TABLE I

Pay for Luck for Oil CEOs (Luck Measure is log Price of Crude Oil)

Dependent Variable: ln (Total Compensation)

Specification:	General (1)	Luck (2)	General (3)	Luck (4)
Accounting rate of return	.82	2.15	—	—
	(.16)	(1.04)		
Log (shareholder wealth)	—	—	.38	.35
			(.03)	(.17)
Age	.05	.07	.05	.05
	(.02)	(.03)	(.02)	(.02)
$\text{Age}^2 * 100$	−.04	−.05	−.04	−.04
	(.02)	(.02)	(.02)	(.02)
Tenure	.01	.01	.01	.01
	(.01)	(.01)	(.01)	(.01)
$\text{Tenure}^2 * 100$	−.03	−.03	−.03	−.03
	(.02)	(.01)	(.02)	(.02)
Firm fixed effects	Yes	Yes	Yes	Yes
Year quadratic	Yes	Yes	Yes	Yes
Sample size	827	827	827	827
Adjusted R^2	.70	—	.75	—

a. Dependent variable is the logarithm of total compensation. Performance measure is accounting rate of return in columns (1) and (2) and the logarithm of shareholder wealth in columns (3) and (4). All nominal variables are expressed in 1977 dollars.
b. Summary statistics for the sample of oil firms are available in Appendix 1.
c. The luck regression (columns (2) and (4)) instrument for performance with the logarithm of the price of a barrel of crude oil in that year, expressed in 1977 dollars.
d. Each regression includes firm fixed effects and a quadratic in year.
e. Standard errors are in parentheses.

TABLE II

Summary Statistics: Full Yermack CEO Sample

	Mean	S. D.
Age of CEO	57.42	6.84
Tenure of CEO	9.10	8.08
Salary and bonus	901.69	795.15
ln (Salary and bonus)	6.62	.60
Total compensation	1595.85	3488.32
ln (Total compensation)	6.98	.81
Number of large shareholders (All)	1.12	1.42
Number of large shareholders on board	.24	.74
Board size	13.45	4.54
Fraction of insiders on board	.42	.19

a. Sample period is 1984–1991.
b. All nominal variables are expressed in thousands of 1991 dollars.
c. Column 1 is the mean for each variable, while column 2 is the standard deviation.
d. A large shareholder is defined as someone who owns more than 5 percent of the common shares in the company, excluding the CEO. Insiders here denote directors who are current or former officers of the company, relatives of corporate officers, or anyone who has a substantial business relationship with the company. Relationships arising in the normal course of business would not be called an insider. Our insider definition corresponds to insider + gray in the original Yermack data.
e. Total compensation equals salary plus bonus plus total value of options grants plus other compensation.

Detailed interpretation. This table gives the scale of compensation and governance variation used in later tests. It also shows why options matter but why the paper focuses on pay components that boards can adjust.

Economic message. The sample contains meaningful variation in both pay and governance, allowing the authors to test whether monitoring changes pay for luck.

7.4 Table III: Pay for luck

Read: Table III estimates pay for luck using exchange-rate shocks and mean industry performance. Pay responds to luck about as much as it responds to general performance. The pattern holds for accounting and market performance measures.

Detailed interpretation. This table is the paper's broad pay-for-luck result. It shows that the oil result generalizes. The industry-performance panel is especially important because it uses a large sample and later becomes the basis for the governance tests.

Economic message. CEO pay-performance sensitivity includes luck sensitivity, not only incentive sensitivity.

7.5 Table IV: Large shareholders and pay for luck

Read: Table IV interacts performance with governance. Large shareholders, especially large shareholders on the board, reduce pay for luck more than they reduce pay for general performance. The strongest reduction is for large shareholders on the board.

Detailed interpretation. The table is central for distinguishing skimming from optimal contracting stories. If paying for luck were optimal, stronger monitors would likely preserve it. Instead, stronger monitors filter it more. This supports the idea that pay for luck reflects CEO influence over the pay process.

TABLE III
PAY FOR LUCK

Dep. var.:	Cash comp		ln (cash)		ln (total comp)		ln (cash)		ln (total comp)	
Specification:	General	Luck	General	Luck	General	Luck	General	Luck	General	Luck
Panel A: Luck Measure is Exchange Rate Shock										
Income	.17	.35	—	—	—	—	—	—	—	—
	(.02)	(.16)								
Income / assets	—	—	2.13	2.94	2.36	4.39	—	—	—	—
			(.16)	(1.28)	(.28)	(2.17)				
ln (shareholder wealth)	—	—	—	—	—	—	.22	.32	.31	.57
							(.02)	(.13)	(.03)	(.23)
Sample size	1737	1737	1729	1729	1722	1722	1713	1713	1706	1706
Adjusted R^2	.75	—	.75	—	.58	—	.75	—	.59	—
Panel B: Luck Measure is Mean Industry Performance										
Income	.21	.34	—	—	—	—	—	—	—	—
	(.02)	(.10)								
Income / assets	—	—	2.18	4.02	2.07	4.00	—	—	—	—
			(.12)	(.53)	(.21)	(.86)				
ln (shareholder wealth)	—	—	—	—	—	—	.20	.22	.25	.29
							(.01)	(.12)	(.02)	(.19)
Sample size	4684	4684	4648	4648	4624	4624	4608	4608	4584	4584
Adjusted R^2	.77	—	.81	—	.70	—	.82	—	.71	—

a. Dependent variable is the level of salary and bonus in columns 1 and 2, the logarithm of salary and bonus in columns 3, 4, 7, and 8 and the logarithm of total compensation in columns 5, 6, 9, and 10. Performance measure is operating income before extraordinary items in columns 1 and 2 (in millions), operating income to total assets in columns 3 to 6 and the logarithm of shareholder wealth in columns 7 to 10. All nominal variables are expressed in real dollars.

b. In the luck regressions in Panel A, the performance measure is instrumented with current and lagged appreciation and depreciation dummies and current and lagged exchange rate index growth. First-stage regressions are presented in Appendix 2.

c. In the luck regressions in Panel B, the performance measure is instrumented with the total assets-weighted average performance measure in the firm's two-digit industry (the firm itself is excluded from the mean calculation).

d. Each regression includes firm fixed effects, year fixed effect and demographic controls (quadratics in age and tenure).

e. Standard errors are in parentheses.

TABLE IV
LARGE SHAREHOLDERS AND PAY FOR LUCK (LUCK MEASURE IS MEAN INDUSTRY PERFORMANCE) *DEPENDENT VARIABLE:* ln (Total Compensation)

Governance measure:	Large shareholders				Large shareholders on board			
Specification:	General (1)	Luck (2)	General (3)	Luck (4)	General (5)	Luck (6)	General (7)	Luck (8)
Income Assets	2.18	4.59	—	—	2.14	4.49	—	—
	(.238)	(.912)			(.217)	(.882)		
Governance* Income/assets	−.094	−.416	—	—	−.181	−1.48	—	—
	(.094)	(.204)			(.176)	(.396)		
ln (shareholder wealth)	—	—	.249	.383	—	—	.258	.318
			(.018)	(.219)			(.017)	(.199)
Governance* Log (shareholder wealth)	—	—	.001	−.066	—	—	−.019	−.076
			(.007)	(.036)			(.016)	(.053)
Governance	−.009	.018	−.017	.411	−.006	.084	.100	.480
	(.011)	(.018)	(.049)	(.240)	(.021)	(.033)	(.108)	(.356)
Sample size	4610	4610	4570	4570	4621	4621	4581	4581
Adjusted R^2	.695		.706		.694		.706	

a. Dependent variable is the logarithm of total compensation. performance measure is operating income to total assets. All nominal variables are expressed in real dollars.

b. In all the luck regressions, both the performance measure and the interaction of the performance measure with the governance measure are instrumented. The instruments are the asset-weighted average performance in the two-digit industry and the interactions of the industry performance with that governance measure.

c. "Large shareholders" indicates the number of blocks of at least 5 percent of the firm's common shares, whether the block holder is or is not a director. "Large shareholders on board" indicates the number of blocks of at least 5 percent of the firm's common shares that are held by directors of the board.

d. Each regression includes firm fixed effects, year fixed effects, a quadratic in age, and a quadratic in tenure.

e. Standard errors are in parentheses.

Economic message. Principals matter: firms with real monitors are better at removing luck from CEO pay.

7.6 Table V: Tenure, large shareholders, and pay for luck

Read: Table V examines CEO tenure. In firms without a large shareholder on the board, tenure increases pay for luck. In firms with a large shareholder on the board, tenure does not have the same effect.

Detailed interpretation. This supports the entrenchment channel. Long tenure can allow CEOs to gain influence over boards. A large shareholder on the board limits that influence and prevents pay for luck from rising with tenure.

Economic message. Tenure becomes problematic when monitoring is weak.

7.7 Table VI: Corporate governance and pay for luck

Read: Table VI uses board size, fraction insiders, and a governance index. The results are less uniformly strong than the large-shareholder results, but the broad pattern supports the idea that better governance reduces pay for luck.

Detailed interpretation. Board-size and insider-share variables are noisier proxies for monitoring than blockholders on the board. The governance-index results show that the pay-for-luck reduction is not purely a one-variable artifact.

Economic message. Governance quality affects the filtering of luck, though the most powerful governance measure is the presence of a large shareholder on the board.

TABLE V

TENURE, LARGE SHAREHOLDERS, AND PAY FOR LUCK (LUCK MEASURE IS MEAN INDUSTRY PERFORMANCE) *DEPENDENT VARIABLE:* ln (TOTAL COMPENSATION)

	Any large shareholder on the board?							
	No		**Yes**		**No**		**Yes**	
Specification:	General (1)	Luck (2)	General (3)	Luck (4)	General (5)	Luck (6)	General (7)	Luck (8)
Income Assets	2.14	3.35	1.28	2.47	—	—	—	—
	(.30)	(.96)	(.65)	(2.60)				
CEO tenure * Income/assets	.00	.13	.063	−.006	—	—	—	—
	(.02)	(.05)	(.045)	(.131)				
Log (shareholder wealth)	—	—	—	—	.24	.26	.27	.53
					(.02)	(.24)	(.05)	(.32)
CEO tenure * Log (sh. wealth)	—	—	—	—	.003	.009	−.005	−.013
					(.001)	(.005)	(.003)	(.010)
CEO tenure	.01	−.00	.010	.016	−.002	−.045	.044	.084
	(.00)	(.01)	(.011)	(.016)	(.01)	(.04)	(.020)	(.059)
Sample size	3884	3884	740	740	3841	3841	743	743
Adjusted R^2	.7030		.757		.715		.700	

a. Dependent variable is the logarithm of total compensation. All nominal variables are expressed in real dollars.

b. In all the luck regressions, both the performance measure and the interaction of the performance measure with the CEO tenure are instrumented. The instruments are the asset-weighted average performance in the two-digit industry and the interactions of the industry performance with the CEO tenure.

c. Sample in columns (1), (2), (5), and (6) is the set of firm-year observations for which there is no large shareholder sitting on the board of directors; sample in columns (3), (4), (7), and (8) is the set of firm-year observations for which there is at least one large shareholder sitting on the board of directors.

d. Each regression includes firm fixed effects, year fixed effects, a quadratic in age, and a quadratic in tenure.

e. Standard errors are in parentheses.

TABLE VI

CORPORATE GOVERNANCE AND PAY FOR LUCK (LUCK MEASURE IS MEAN INDUSTRY PERFORMANCE) *DEPENDENT VARIABLE:* ln (TOTAL COMPENSATION)

Governance measure:	Board size				Fraction insiders			
Specification:	General (1)	Luck (2)	General (3)	Luck (4)	General (5)	Luck (6)	General (7)	Luck (8)
Income Assets	2.61	5.19	—	—	2.30	2.27	—	—
	(.558)	(1.62)			(.453)	(1.24)		
Governance* Income/assets	−.045	−.093	—	—	−.482	4.51	—	—
	(.043)	(.094)			(.853)	(2.69)		
Log (shareholder wealth)	—	—	.216	.099	—	—	.241	.241
			(.034)	(.210)			(.029)	(.215)
Governance* Log (sh. wealth)	—	—	.002	.013	—	—	.027	.126
			(.002)	(.006)			(.05)	(.190)
Governance	.012	.015	−.013	−.080	.158	−.315	−.066	−.742
	(.005)	(.007)	(.016)	(.041)	(.129)	(.271)	(.407)	(1.29)
Sample size	4624	4624	4584	4584	4624	4624	4584	4584
Adjusted R^2	.695		.706		.695		.706	

Governance measure:	Governance index			
Specification:	General (9)	Luck (10)	General (11)	Luck (12)
Income Assets	2.07	4.23	—	—
	(.210)	(.865)		
Governance* Income/assets	.007	−.216	—	—
	(.057)	(.134)		
Log (shareholder wealth)	—	—	.249	.252
			(.016)	(.232)
Governance* Log (sh. wealth)	—	—	−.003	−.033
			(.004)	(.015)
Governance	−.016	.000	.010	.210
	(.007)	(.011)	(.027)	(.103)
Sample size	4610	4610	4551	4551
Adjusted R^2	.695		.705	

a. Dependent variable is the logarithm of total compensation. All nominal variables are deflated. Each regression includes firm fixed effects, year fixed effects, a quadratic in age, and a quadratic in tenure. Standard errors are in parentheses.

b. "Board size" indicates the number of members of the board of directors, as listed in the proxy statement near the start of the fiscal year. "Fraction insiders" is the fraction of inside and "gray" directors on the board of directors. "Governance index" is the unweighted average of four standardized governance variables (number of large shareholders, number of large shareholders on board, minus board size, and one minus fraction insiders).

c. In all the luck regressions, both the performance measure and the interaction of the performance measure with the governance measure are instrumented. The instruments are the asset-weighted average performance in the two-digit industry and the interactions of the industry performance with that governance measure.

TABLE VII
CORPORATE GOVERNANCE AND PAY FOR LUCK ROBUSTNESS CHECKS
(LUCK MEASURE IS MEAN INDUSTRY PERFORMANCE)
DEPENDENT VARIABLE: ln (TOTAL COMPENSATION)

Governance measure:	Large shareholders on board				Governance index			
Specification:	General (1)	Luck (2)	General (3)	Luck (4)	General (5)	Luck (6)	General (7)	Luck (8)
Income Assets	.288	36.9	—	—	−.570	12.3	—	—
	(1.14)	(14.6)			(1.31)	(6.17)		
Governance* Income/assets	−.118	−2.23	—	—	.056	−.197	—	—
	(.18)	(.52)			(.061)	(.126)		
Log (shareholder wealth)	—	—	.223	−.136	—	—	.251	−.194
			(.086)	(.334)			(.098)	(.345)
Governance* Log (shareholder wealth)	—	—	−.018	−.059	—	—	−.003	−.027
			(.016)	(.053)			(.004)	(.012)
Governance	−.010	.127	.094	.365	−.019	−.002	.007	.170
	(.021)	(.038)	(.109)	(.357)	(.007)	(.010)	(.029)	(.085)
Firm Size ∗ Performance?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample size	4621	4621	4581	4581	4610	4610	4551	4551
Adjusted R^2	.695	—	.706	—	.695	—	.705	—

a. Dependent variable is the logarithm of total compensation. Performance measure is operating income to total assets. All nominal variables are expressed in real dollars. Each regression includes firm fixed effects, year fixed effects, a quadratic in age, and a quadratic in tenure.

b. In all the luck regressions, both the performance measure and the interaction of the performance measure with the governance measure are instrumented. The instruments are the asset-weighted average performance in the two-digit industry and the interactions of the industry performance with that governance measure.

c. "Large shareholders on board" indicates the number of blocks of at least 5 percent of the firm's common shares that are held by directors of the board. "Governance index" is the unweighted average of four standardized governance variables (number of large shareholders, number of large shareholders on board, minus board size, and one minus fraction insiders). "Firm size" indicates average log assets over the same period.

d. Standard errors are in parentheses.

7.8 Table VII: Corporate governance and pay for luck robustness checks

Read: Table VII adds robustness checks that interact firm size with performance. The main governance results survive these controls. Large shareholders on the board continue to reduce pay for luck.

Detailed interpretation. The table addresses the concern that governance measures simply proxy for firm size. By allowing pay-performance sensitivity to vary with firm size, the paper shows that the governance effect is not just a size effect.

Economic message. The governance evidence remains after controlling for an important alternative explanation.

8 Short summary

- The paper tests whether CEO pay filters out observable luck.
- Simple contracting theory predicts that boards should not reward CEOs for luck that shareholders can observe.
- The paper studies oil prices, exchange-rate shocks, and industry mean performance as luck measures.
- CEO pay rises with luck in all three settings.
- Pay for luck is roughly as large as pay for general performance.
- Pay for luck appears even in salary and bonus, not only in mechanically unindexed option values.
- Governance matters: firms with large shareholders and stronger boards pay less for luck.
- Large shareholders on the board reduce pay for luck by about 23–33 percent.
- CEO tenure increases pay for luck when there is no large shareholder on the board, consistent with entrenchment.

- The findings support a skimming view in poorly governed firms and a contracting view in better governed firms.

9 Conclusion

9.1 Core conclusion

CEOs are rewarded for observable luck. This is inconsistent with the simplest contracting benchmark, which predicts that observable luck should be filtered from pay. The effect is not small: pay responds to lucky performance about as much as it responds to general performance.

9.2 Economic intuition

Under optimal contracting, observable luck should be removed because it adds risk without improving incentives. Under skimming, however, good performance creates slack. When the firm is doing well, shareholders scrutinize pay less, allowing CEOs to extract more pay even if the performance improvement is lucky.

9.3 Why governance matters

The governance tests are the paper's key contribution beyond documenting pay for luck. Better governed firms, especially those with large shareholders on the board, pay less for luck. This weakens explanations based on CEO outside options or inability to filter luck. If filtering were impossible, well-governed firms could not do it better.

9.4 Implications for CEO pay research

The paper warns that pay-performance sensitivity can be misleading. A CEO can have high pay-performance sensitivity because pay rises with luck rather than effort. Researchers must distinguish pay for performance from pay for the component of performance outside managerial control.

9.5 Limitations

The governance results are cross-sectional and not a randomized governance reform. The luck measures may not be perfect, and extended contracting models can rationalize some pay for luck. Still, the governance interaction evidence is difficult to reconcile with purely optimal pay-for-luck stories.

9.6 Takeaway

Principal-agent models explain CEO pay best when there are real principals. Poorly governed firms look more like skimming environments, while well-governed firms do more to filter luck from pay.